

E-commerce Warehouse Location Optimization for E-Commerce in India

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1 Problem

1.1 Research Problem

The explosive growth of e-commerce in India, projected to reach \$163 billion by 2026 with a CAGR of 21.5% The Hindu Businessline, has intensified competition among online retailers. In this hypercompetitive landscape, efficient logistics operations have emerged as a critical differentiator. The “last mile” delivery problem is particularly challenging in India’s urban centers, where poor warehouse placement decisions lead to extended delivery times, increased fuel consumption, and ultimately, lost market share and customer dissatisfaction.

Traditional approaches to warehouse location often rely on heuristics or rules of thumb rather than mathematical optimization, resulting in suboptimal facility networks that increase operational costs. Furthermore, most existing models fail to account for the inherently stochastic nature of customer demand, leading to capacity imbalances and service failures during demand fluctuations.

This study addresses how to optimally select warehouse locations from a set of candidate sites to minimize total fixed and transportation costs, while accounting for random demand. Our approach brings together operations research techniques and statistical modeling to develop a decision support framework for e-commerce logistics planning.

1.2 Research Question

How can we minimize the cost of serving stochastic customer demand by optimally choosing warehouse locations from a finite set of candidate sites?

2 Plan

We model this challenge as an uncapacitated facility location problem (FLP). Our approach utilizes Mixed Integer Linear Programming (MILP) with stochastic elements to determine the optimal subset of warehouses to open.

What to measure We consider three key variables:

1. Demand volume at customer locations,
2. Coordinates of customers and candidate warehouses,
3. Transportation costs between facilities and demand points.

How to measure Demands are modeled as random variables drawn from some distributions, reflecting the unpredictability of the e-Commerce order patterns. Transportation costs are calculated via Euclidean distance, which serves as a proxy for actual delivery distances in urban environments.

Study design We simulate a metropolitan area as a square region $[0, 100]^2$ units containing 50 customer demand points and 7 candidate warehouse locations.

Optimization technique Our MILP formulation incorporates:

- Demand satisfaction constraints,
- Capacity limitations,
- Linking constraints that connect facility opening decisions to customer assignments.

3 Data

3.1 Data Description

Given the proprietary nature of actual e-commerce demand and the sensitivity of real estate information for potential warehouse sites, we generate synthetic data that capture the essential characteristics of the problem.

- **Customer Demand Points:** 50 observations, each with spatial coordinates (x, y) and an expected demand

$$\lambda_i \sim \text{Uniform}(50, 200) \quad \text{for } i = 1, \dots, 50.$$

- **Warehouse Candidates:** 7 potential locations with approximately identical fixed setup cost of 1,000 units each, representing the annualized cost of leasing, staffing, and operating a mid-sized distribution center.

3.2 Variable Summary

Variable	Type	Description
x, y	Numeric	Spatial coordinates of points
λ	Numeric	Expected demand at each customer location
c_{ij}	Numeric	Distance-based transport cost between i and j
y_j	Binary	Indicator: warehouse j open (1) or closed (0)
x_{ij}	Continuous	Amount of demand i served by warehouse j

3.3 Data Handling

The data generation process follows these mathematical steps:

1. Sample demand locations uniformly throughout the region:

$$x_i \sim \text{Uniform}(0, 100), \quad y_i \sim \text{Uniform}(0, 100), \quad i = 1, \dots, 50.$$

2. Sample warehouse candidate coordinates similarly:

$$x_j \sim \text{Uniform}(0, 100), \quad y_j \sim \text{Uniform}(0, 100), \quad j = 1, \dots, 5.$$

3. Generate expected demand for each customer point:

$$\lambda_i \sim \text{Uniform}(50, 200), \quad i = 1, \dots, 50.$$

4. Distance:

$$c_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}, \quad i = 1, \dots, 50, j = 1, \dots, 5.$$

This procedure yields a spatial distribution of demand points over a 100×100 unit² area with heterogeneous demand levels and candidate warehouse sites.

4 Analysis

4.1 Optimization Formulation

The mathematical formulation of our Facility Location Problem (FLP) is as follows:

Sets

- I : Set of customer locations, indexed by i
- J : Set of potential warehouse locations, indexed by j

Parameters

- λ_i : Expected demand at customer i
- c_{ij} : Transportation cost per unit from customer i to warehouse j
- F_j : Fixed cost of opening warehouse j
- M_j : A large constant, representing warehouse j 's capacity (effectively uncapacitated)

Decision Variables

- $y_j \in \{0, 1\}$: 1 if warehouse j is opened, 0 otherwise
- $x_{ij} \geq 0$: Units of demand at customer i satisfied by warehouse j

Objective Function Minimize total cost:

$$\min \sum_{j \in J} F_j y_j + \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij}$$

Subject to constraints:

- **Demand satisfaction:**

$$\sum_{j \in J} x_{ij} = \lambda_i, \quad \forall i \in I$$

- **Capacity (Big-M) constraints:**

$$\sum_{i \in I} x_{ij} \leq M_j y_j, \quad \forall j \in J$$

- **Strong linking constraints:**

$$x_{ij} \leq \lambda_i y_j, \quad \forall i \in I, j \in J$$

- **Variable domains:**

$$y_j \in \{0, 1\}, \quad x_{ij} \geq 0$$

This model was implemented and solved using the `ompr` and `R0I.plugin.symphony` packages in R and the full implementation of this model can be accessed at: Google Colab Notebook.

4.2 Solution and Results

For a particular generated setup that was done in R:

- **Optimal cost:** 4,753.28 (in scaled units)
- **Warehouses opened:** 2

The solution indicates that only 2 out of 5 potential warehouses should be opened to minimize total cost while satisfying all customer demand. The optimal cost incorporates both the fixed setup costs and the variable transportation costs.

Benchmark Comparison

- **Naive approach (open all warehouses):** 5,730.15 (in scaled units)
- **Cost savings from optimization:** 17.047%

This comparison demonstrates the value of mathematical programming in reducing logistics costs by nearly 17% compared to a heuristic strategy.

4.3 Visualization

To better understand the spatial implications of our solution, we visualized the customer–warehouse assignments and the geographical distribution of facilities (for yet another generated data sets):

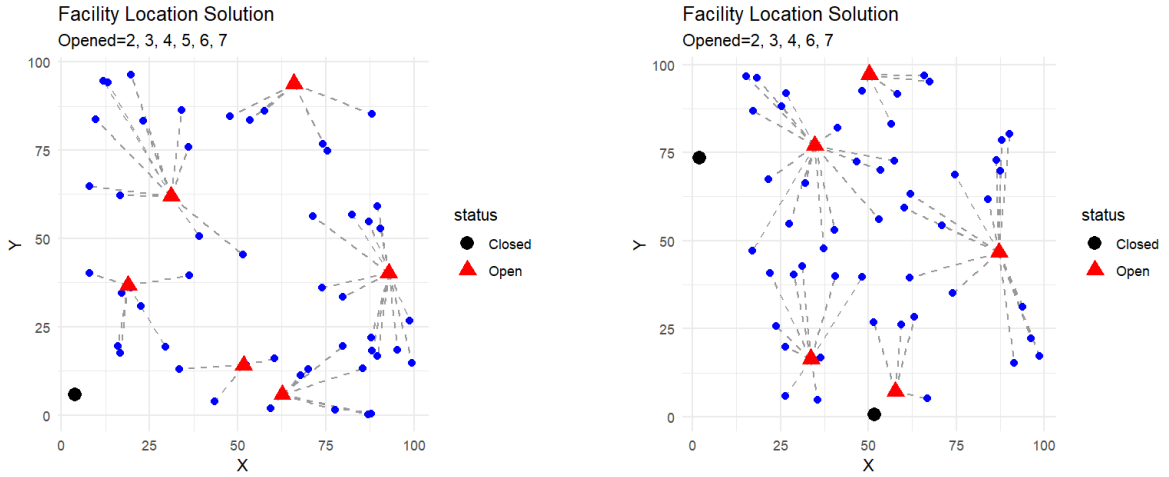


Figure 1: Optimal warehouse locations (red triangles) and customer assignments (blue dots). Dashed lines indicate assignment of each customer to its serving facility.

The visualization reveals how the optimization model strategically placed warehouses to minimize overall system costs. The opened facilities are positioned to efficiently serve clusters of customer demand, with assignments based on proximity and demand volume.

5 Conclusion and Communication

5.1 Key Findings

Our analysis of the e-commerce warehouse location problem yielded several important insights:

- **Selective Facility Opening:** The optimal solution required opening only fewer potential warehouses, highlighting the inefficiency of establishing excessive distribution points.
- **Stochastic Consideration:** By incorporating expected demand values drawn from probability distributions, our model provides robustness against typical demand variations.
- **Cost Efficiency:** The optimized solution achieved some significant cost savings compared to a naive approach, translating to significant operational advantages in the competitive e-commerce landscape.
- **Demand-Driven Assignments:** The model effectively balanced transportation costs against fixed facility costs, creating an efficient assignment pattern that accounts for the spatial distribution of customer demand.

5.2 Limitations

Despite the strengths of our approach, several limitations should be acknowledged:

- **Distance Metric:** The use of Euclidean distance is a simplification of actual transportation networks. In urban environments, road networks with varying congestion levels would significantly impact delivery times and costs.
- **Homogeneous Facilities:** Our model assumed identical fixed costs and unlimited capacity for all potential warehouses, whereas real estate costs and physical constraints vary by location.
- **Demand Distribution:** While we modeled demand as uniformly distributed, actual e-commerce order patterns likely follow more complex distributions with temporal variations (e.g., seasonality, time-of-day effects).
- **Single-Period Model:** The current model takes a static view, whereas warehouse location decisions are inherently strategic with multi-period implications.

5.3 Future Work

Several promising directions for future research emerge from this study:

- **Network-Based Distance:** Incorporate actual road network data to calculate more realistic transportation costs based on vehicle routing problem—VRP algorithms.
- **Multi-Period Modeling:** Extend the model to capture demand evolution over time and allow phased facility openings.

- **Empirical Demand Patterns:** Replace synthetic demand with actual order data from e-commerce platforms to better capture spatial and temporal patterns.
- **Service Level Constraints:** Add constraints ensuring time-based delivery promises (e.g., same-day or next-day delivery) can be met from the selected warehouses.
- **Multi-Echelon Extensions:** Expand the model to include multiple tiers of facilities (e.g., fulfillment centers, dark stores, and micro-fulfillment centers).

In conclusion, this study demonstrates the value of mathematical optimization for strategic logistics decisions in e-commerce. The MILP approach with stochastic demand consideration provides a powerful framework that can be extended to address the complex realities of urban logistics networks, ultimately contributing to more efficient and competitive e-commerce operations.

References

1. Optimal Facility Location – Wikipedia.
2. SCIP Optimization Suite Documentation.